

# Om Patel

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## EDUCATION

**California State University, Long Beach**  
*Bachelor of Science in Computer Science*

Jan 2024 – Dec 2027  
Long Beach, CA

## EXPERIENCE

### Undergraduate Research Assistant

May 2026 – Present

*California State University, Long Beach*

*Long Beach, CA*

- Built multithreaded **Java/C++** programs studying **parallel execution**, optimizing CPU-bound workloads across thread pools by tuning task-scheduling and workload-partitioning strategies.
- Developing an AI agent on a **C#/.NET** backend integrated with **ServiceNow** that classifies, triages, and auto-resolves incident and service-request tickets, on track to reduce organization-wide ticket volume by **30%**.
- Built live dashboards over the **ServiceNow REST API** surfacing ticket trends, recurring failure modes, and high-volume categories, now used to prioritize which resolution workflows to automate first.
- Designed **confidence-based routing** that resolves tickets only on strong knowledge-base retrieval and escalates ambiguous cases to human agents, scaling automation without sacrificing resolution accuracy.

### Software Engineer Intern

July 2025 – Present

*Beach Building Services (CSULB)*

*Long Beach, CA*

- Engineered and integrated a **Gemini LLM** feature into a production web application using structured prompt pipelines, input validation, and rate-aware fallbacks, sustaining **100+ concurrent requests/min** with **zero downtime**.
- Designed and implemented a normalized **PostgreSQL** relational schema with analytics views, indexed lookups, and aggregation queries, powering stakeholder reporting across professor and course rating data.
- Optimized performance-critical **C/C++** data parsing and validation utilities on **Linux**, reducing processing time by **30%** via memory-efficient algorithms and low-overhead I/O.

### AI Engineer Intern

September 2025 – December 2025

*Impacter AI*

*Remote*

- Built and deployed **Python REST API** microservices with **ETL** data-transformation pipelines that integrated LLM inference into production, reducing end-to-end latency by **180ms** through async processing and response caching.
- Resolved concurrent-request **race conditions** and improved LLM output accuracy by **12%** using structured output schemas, post-processing validation, and systematic prompt engineering.
- Fine-tuned **GPT-3** and **Llama 2** with few-shot prompting and quantization, cutting compute costs by **35%** and average token usage by **28%** across production workloads.

## PROJECTS

**RepoSage AI** | *Python, FastAPI, LangChain, PostgreSQL, pgvector, OpenAI API, GitHub API, Redis, BullMQ, Docker*

May 2026

- Engineered an **agentic RAG** system over GitHub repositories using **LangChain** tool-calling, enabling multi-step code understanding, architecture explanation, and bug detection across **50k+ line** codebases.
- Orchestrated an async distributed embedding pipeline with **BullMQ** and **pgvector**, applying semantic chunking strategies to improve retrieval precision by **38%** over naive fixed-size approaches.
- Generated structured onboarding docs, refactor suggestions, and architectural summaries via chained LLM prompts grounded in retrieved context, surfacing actionable insights in **under 5 seconds**.

**SimmSilos** | *Python, FastAPI, TypeScript, PostgreSQL, Redis, Docker, Apache*

May 2026

- Architected a **multi-service platform** behind an **Apache gateway**: a **FastAPI** service for auth, branch/task assignment, and silo file access, sharing state across services through **PostgreSQL + Redis**.
- Built a **TypeScript VS Code extension** implementing a custom **virtual filesystem** with full save support, so developers browse, edit, and write files inside isolated silos alongside live task and branch sidebars.
- Engineered ephemeral developer environments: a **port registry** assigns each developer a dedicated **Docker container** on a unique port; on branch submission the container is destroyed, logs persisted for audit, and the port freed for the next developer.

**DevSecOps Project** | *Jenkins, Docker, Kubernetes, Helm, ArgoCD, SonarQube, Trivy, Prometheus, Grafana, AWS*

Feb 2025

- Designed a **Jenkins pipeline** with **Docker build/scan** (**SonarQube, Trivy**), blocking **15+ vulnerabilities** before deployment.
- Automated deployments with **Helm + ArgoCD** on **Kubernetes**, streamlining **GitOps workflows** and improving delivery speed.
- Integrated **Prometheus** and **Grafana**, enhancing system monitoring and reliability for **production use**.

## TECHNICAL SKILLS

**Languages:** Python, SQL, C, C++, Go, Java, JavaScript, TypeScript

**AI/ML & Data:** PyTorch, LangChain, RAG, Embeddings, Agents, Fine-tuning, Prompt Engineering, Airflow, dbt, Kafka, Spark, ETL/ELT, pandas, NumPy, OpenCV, MLflow

**Backend & Storage:** FastAPI, Node.js, Flask, PostgreSQL, pgvector, BigQuery, MySQL, MongoDB, Redis, AWS, Google Cloud, Firebase

**Systems & Tools:** Linux, Git, Docker, Kubernetes, CI/CD, GitHub Actions, REST APIs, WebSockets, Prometheus, Grafana, LangSmith, Agile